



**ServiFinance: A Multi-Tenant SaaS Service Management System with
Embedded Micro-Lending**

**In Partial fulfillment of the requirements
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Business Case

Background

ServiFinance is a multi-tenant service management and finance platform. Its web side, called SMS, handles tenant service operations such as customer records, service requests, dispatch, technician work, status tracking, invoicing, and reports. Its desktop side, called MLS, handles finance workflows that should be separated from normal service operations, including invoice-to-loan conversion, standalone loan processing, amortization, payment posting, collections, ledger review, audit review, and financial reporting.

Purpose

The purpose of ServiFinance is to address the operational and financial challenges faced by Philippine MSMEs by integrating service management and micro-lending into a single secure system. Many businesses can deliver high-value services but struggle to provide structured and reliable credit options to customers. ServiFinance bridges this gap by enabling enterprises to manage services, customer accounts, and micro-loans within one platform, improving both operational efficiency and financial control.

Identified Problems and Proposed Solutions

Problem 1: Limited ability of MSMEs to offer structured credit

Many MSMEs lose potential sales because they cannot provide organized installment-based payment options for high-cost services. Access to financing remains a persistent constraint for small businesses, limiting growth opportunities and customer retention (IFC, 2017).

Solution:

ServiFinance provides an embedded micro-lending module that allows businesses to convert completed services into manageable credit transactions. This helps MSMEs offer flexible payment arrangements while maintaining proper monitoring of loans, due dates, and balances.

Problem 2: Manual financial recording leads to transaction errors

Businesses that rely on handwritten ledgers or spreadsheets for loan tracking are exposed to errors in interest computation, balance updates, and payment reconciliation. Spreadsheet-based processes are especially prone to hidden formula and encoding mistakes, which can undermine financial accuracy (Panko, 2016).

Solution:

The desktop terminal automates loan computation, amortization schedules, interest calculations, and payment posting. This reduces human error and ensures more accurate, consistent, and traceable financial records.

Problem 3: Disconnected records and weak security create data risks

When service records, customer information, and payment histories are stored separately, businesses face poor data visibility, delayed reconciliation, and reduced decision quality. At the same time, sensitive financial information requires controlled access and proper monitoring to reduce misuse and unauthorized handling (Caserio & Trucco, 2020; NIST, 2024).

Solution:

ServiFinance centralizes service, customer, and lending data in one secure environment with role-based access and audit logging. This improves data integrity, strengthens accountability, and supports safer management of sensitive financial transactions.

Objectives

- Provide a secure desktop entry point for tenant finance users.
- Convert finalized SMS invoices into micro-loan accounts.
- Support standalone loan creation when the loan does not originate from a service invoice.
- Calculate amortization schedules based on principal, interest rate, term, and start date.
- Post loan payments and update installment status, running balance, and ledger records.
- Provide customer financial records that combine loans and payment history.
- Provide collections, reports, ledger, audit, role, and platform-user administration views.
- Keep SMS service operations and MLS finance operations separated by surface and role scope.

Scope

The Desktop MLS scope includes:

- MLS desktop login
- MLS dashboard
- customer financial records
- invoice-to-loan conversion
- standalone loan processing
- loan accounts and payment posting
- collections queue
- reports
- ledger review
- audit review
- platform users
- roles and permissions

The current desktop MLS is focused on tenant finance operations. Superadmin desktop mode, multi-tenant employee switching, deeper export formats, and more advanced correction workflows can be treated as future enhancements unless required for final submission.

Stakeholders

Stakeholder	Interest In The System
Tenant Owner / Administrator	Controls tenant users, roles, reports, and finance oversight.
Finance Staff / Loan Officer	Converts invoices into loans, creates standalone loans, reviews accounts, and monitors balances.
Cashier	Posts payments and checks collections or ledger entries.
SMS Staff / Dispatcher / Technician	Produces the service and invoice records that may become finance-ready.
Customer	Receives service, invoice, and loan/account settlement visibility through customer-facing records.
Superadmin	Governs the platform, tenant subscriptions, modules, and root-level administration.

Business Benefits

- Reduces duplicate encoding between service operations and finance records.
- Improves traceability from service request to invoice to loan account.
- Helps finance staff compute installment schedules consistently.
- Provides a dedicated ledger trail for loan creation, payment posting, and reversals.
- Separates desktop finance functions from web-based service operations.
- Supports premium tenant value through desktop finance modules.
- Improves auditability for sensitive financial actions.

Risks And Controls

Risk	Control
Incorrect money computation	Use backend-controlled amortization, payment posting, and ledger update rules.
Unauthorized MLS access	Use desktop-only routing, authenticated sessions, role checks, and tenant scope validation.
Duplicate payment posting	Validate action state and record ledger transactions with references.
Weak traceability	Store finance actions in ledger transactions and audit events.
Tenant data leakage	Keep tenant-owned records linked by TenantId and protect APIs by tenant context.
Desktop/backend startup failure	Desktop host bootstraps and waits for the owned backend before using API-backed screens.

Importance

ServiFinance is important because it enables MSMEs to improve customer service, increase revenue opportunities, and maintain stronger control over financial operations. By combining service management and micro-lending in one desktop-based system, it supports

more reliable decision-making, more secure data handling, and more sustainable business growth.

Conclusion

In conclusion, the desktop-side Micro-lending System addresses three major business problems encountered by MSMEs: the limited ability to offer structured credit, the high risk of transaction errors caused by manual recording, and the security and data risks created by disconnected records. To respond to these issues, the proposed system integrates micro-lending functions with service management, automates financial computations such as amortization and payment tracking, and centralizes sensitive records in a controlled desktop environment with improved access management and auditability. These recommendations strengthen operational efficiency, improve financial accuracy, reduce risk, and support better decision-making. Overall, the proposed solution contributes to business improvement by enabling MSMEs to serve customers more effectively while maintaining secure, reliable, and well-organized financial operations.

Methodology

This business case used a descriptive and problem-solution approach in analyzing the needs of MSMEs and in developing the proposed desktop-based Micro-lending System. The study began by identifying common operational and financial difficulties experienced by small businesses, particularly in handling installment payments, monitoring receivables, and maintaining accurate service and payment records. Relevant academic literature, institutional reports, and security guidance were reviewed to establish the practical and theoretical basis of the identified problems.

The analysis focused on how current manual or disconnected processes affect transaction accuracy, information quality, and data security. From these findings, corresponding system-based solutions were formulated using a business process improvement perspective. The proposed features of the desktop system, such as automated loan computation, centralized record management, and controlled user access, were then aligned with the objective of improving operational efficiency and financial reliability. This methodology ensured that the recommendations were not only technically appropriate but also responsive to the actual business challenges faced by MSMEs.

Tools

1. Microsoft Word - Used to write, organize, and format the business case, proposal content, and supporting documentation.

2. Visual Studio Code (VS Code) - Used for editing source code, reviewing project files, managing frontend and backend components, and maintaining documentation during development.

3. Visual Studio 2022/2026 - Used as the primary integrated development environment for building, debugging, and running the .NET-based components of the ServiFinance system, including the backend API and desktop application.

4. .NET and ASP.NET Core - Used to develop the core backend services of the system, including authentication, multi-tenant processing, API endpoints, business logic, and system integration.

5. .NET MAUI Hybrid - Used for the desktop-side implementation of the Micro-lending System, allowing the project to combine desktop application capabilities with a shared web-based user interface.

6. React and TypeScript - Used to build the user interface for the system's web and desktop-facing screens, including dashboards, authentication views, and tenant-facing modules.

7. Vite - Used as the frontend build tool for compiling, bundling, and serving the React and TypeScript application efficiently during development.

8. SQL Server - Used as the primary relational database for storing tenants, users, customers, service requests, invoices, micro-loans, amortization schedules, and ledger transactions.

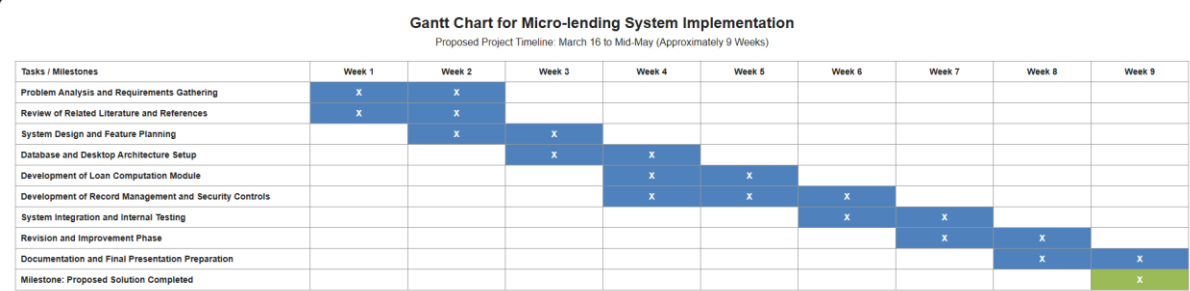
9. SQL Server Management Studio (SSMS) - Used to inspect, manage, and validate database structures, records, and queries during development and testing.

10. Entity Framework Core - Used for data access, database modeling, migrations, and interaction between the application and SQL Server database.

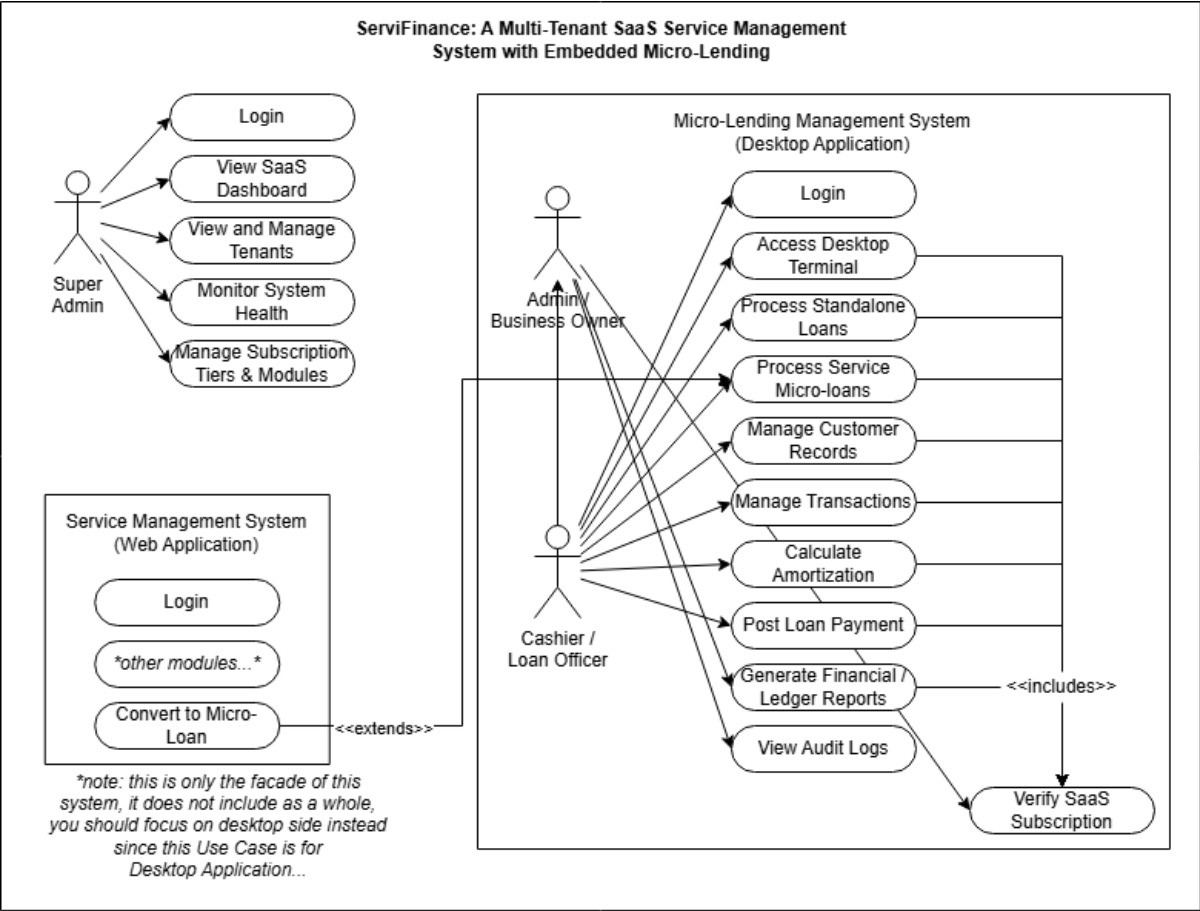
11. Git and GitHub - Used for version control, source code management, and tracking project changes during system development.

12. Online Research Sources and Academic References - Used to gather supporting literature, industry context, and credible citations related to MSME financing, transaction accuracy, ERP integration, and information security.

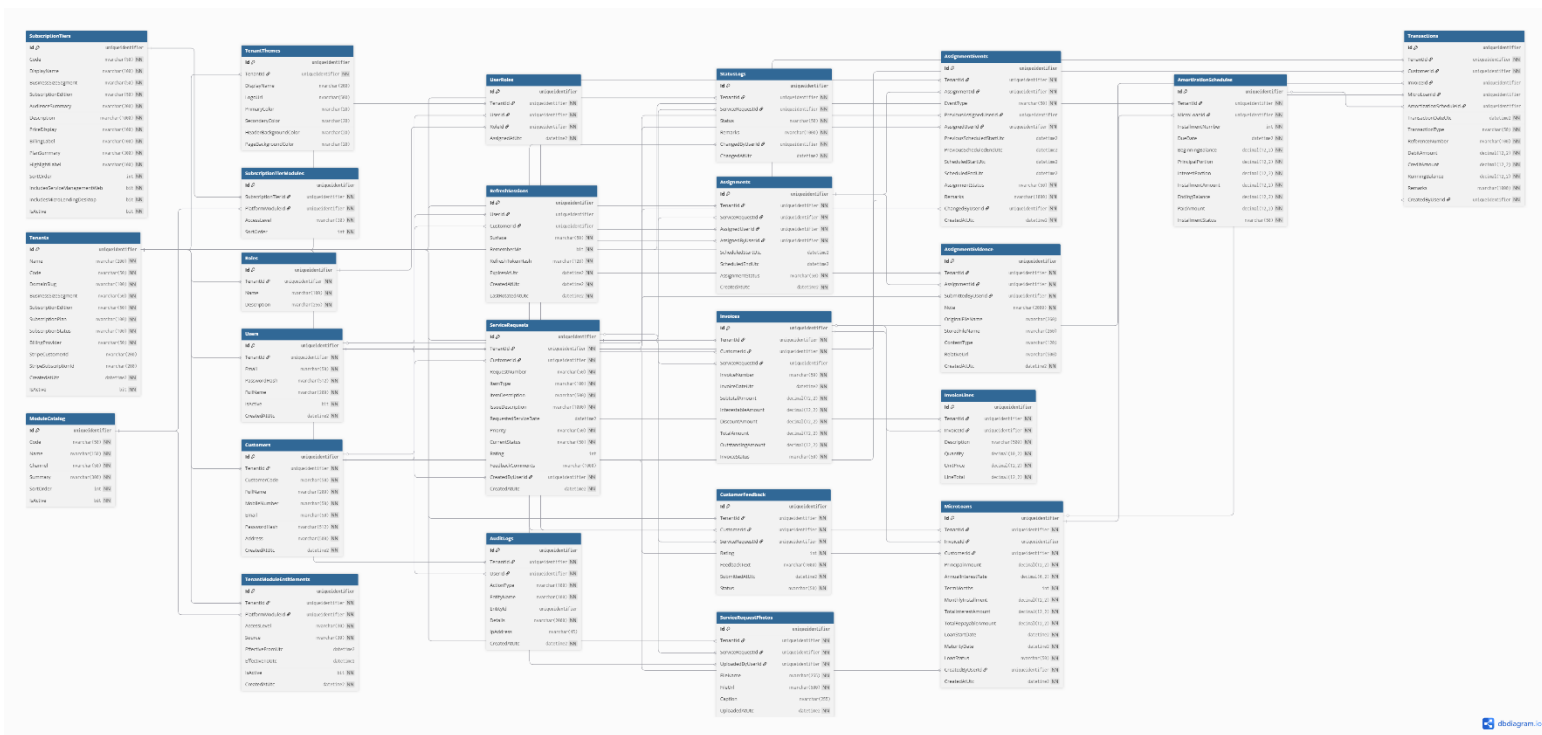
Gantt Chart



Use Case Diagram



Entity Relationship Diagram



Data Dictionary

Tenants

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the tenant.
BusinessName	nvarchar	200	Tenant business name.
DomainSlug	nvarchar	100	Tenant route/domain identifier.
SubscriptionTierId	uniqueidentifier	36	Selected subscription tier.
IsActive	bit	1	Indicates whether the tenant is active.
CreatedAtUtc	datetime2	-	Date and time when the tenant was created.

Users

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the user.
TenantId	uniqueidentifier	36	Tenant that owns the user, when tenant-scoped.
FullName	nvarchar	200	User full name.
Email	nvarchar	50	User email address used for login.
PasswordHash	nvarchar	512	Hashed password value.
IsActive	bit	1	Indicates whether the account can sign in.

CreatedAtUtc	datetime2	-	Date and time when the user was created.
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Roles

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the role.
TenantId	uniqueidentifier	36	Tenant that owns the role, when tenant-scoped.
Name	nvarchar	100	Role name such as Owner, Administrator, SMS Staff, MLS Staff, or MLS Cashier.
PlatformScope	nvarchar	30	Role scope such as Root, OwnerAdmin, SMS, or MLS.
Rank	int	10	Role rank used for ordering and governance.
IsSystemRole	bit	1	Indicates whether the role is system-defined.

UserRoles

Field	Data Type	Length / Precision	Description
UserId	uniqueidentifier	36	Foreign key to Users.Id.
RoleId	uniqueidentifier	36	Foreign key to Roles.Id.

Customers

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the customer.
TenantId	uniqueidentifier	36	Tenant that owns the customer.
CustomerCode	nvarchar	50	Unique customer code within the tenant.
FullName	nvarchar	200	Customer full name.
MobileNumber	nvarchar	50	Customer contact number.
Email	nvarchar	50	Customer email address.
Address	nvarchar	500	Customer address.
CreatedAtUtc	datetime2	-	Date and time when the customer was created.

Invoices

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the invoice.
TenantId	uniqueidentifier	36	Tenant that owns the invoice.
CustomerId	uniqueidentifier	36	Linked customer.

ServiceRequestId	uniqueidentifier	36	Optional source service request.
InvoiceNumber	nvarchar	50	Unique invoice number within the tenant.
InvoiceDateUtc	datetime2	-	Invoice issue date.
SubtotalAmount	decimal	12,2	Total before discounts.
InterestableAmount	decimal	12,2	Amount eligible for financing.
DiscountAmount	decimal	12,2	Discount applied to the invoice.
TotalAmount	decimal	12,2	Final invoice total.
OutstandingAmount	decimal	12,2	Remaining unpaid balance.
InvoiceStatus	nvarchar	50	Current invoice status.

InvoiceLines

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the invoice line.
TenantId	uniqueidentifier	36	Tenant that owns the invoice line.
InvoiceId	uniqueidentifier	36	Linked invoice.
Description	nvarchar	500	Billed item or service description.
Quantity	decimal	10,2	Quantity billed.
UnitPrice	decimal	12,2	Unit price.
LineTotal	decimal	12,2	Line total.

MicroLoans

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the micro-loan.
TenantId	uniqueidentifier	36	Tenant that owns the loan.
InvoiceId	uniqueidentifier	36	Optional source invoice.
CustomerId	uniqueidentifier	36	Linked customer.
PrincipalAmount	decimal	12,2	Principal loan amount.
AnnualInterestRate	decimal	6,2	Annual interest rate.
TermMonths	int	10	Repayment term in months.
MonthlyInstallment	decimal	12,2	Monthly installment amount.
TotalInterestAmount	decimal	12,2	Total interest for the loan.
TotalRepayableAmount	decimal	12,2	Principal plus total interest.
LoanStartDate	datetime2	-	Loan start date.
MaturityDate	datetime2	-	Final due date.
LoanStatus	nvarchar	50	Current loan status.
CreatedByUserId	uniqueidentifier	36	User who created the loan.
CreatedAtUtc	datetime2	-	Date and time when the loan was created.

AmortizationSchedules

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the schedule row.

TenantId	uniqueidentifier	36	Tenant that owns the schedule row.
MicroLoanId	uniqueidentifier	36	Linked micro-loan.
InstallmentNumber	int	10	Sequence number of the installment.
DueDate	datetime2	-	Installment due date.
BeginningBalance	decimal	12,2	Balance before the installment.
PrincipalPortion	decimal	12,2	Principal part of the installment.
InterestPortion	decimal	12,2	Interest part of the installment.
InstallmentAmount	decimal	12,2	Total installment amount.
EndingBalance	decimal	12,2	Balance after the installment.
PaidAmount	decimal	12,2	Amount already paid.
InstallmentStatus	nvarchar	50	Status such as Unpaid, Partial, or Paid.

Transactions

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the ledger transaction.
TenantId	uniqueidentifier	36	Tenant that owns the transaction.
CustomerId	uniqueidentifier	36	Linked customer.
InvoiceId	uniqueidentifier	36	Optional linked invoice.
MicroLoanId	uniqueidentifier	36	Optional linked micro-loan.
AmortizationScheduleId	uniqueidentifier	36	Optional linked amortization row.
ReversalOfTransactionId	uniqueidentifier	36	Optional original transaction when this row is a reversal.
TransactionDateUtc	datetime2	-	Date and time of the transaction.
TransactionType	nvarchar	50	Type such as Loan, LoanPayment, or LoanPaymentReversal.
ReferenceNumber	nvarchar	100	Receipt or ledger reference number.
DebitAmount	decimal	12,2	Debit amount.
CreditAmount	decimal	12,2	Credit amount.
RunningBalance	decimal	12,2	Balance after posting.
Remarks	nvarchar	1000	Transaction notes.
CreatedByUserId	uniqueidentifier	36	User who posted the transaction.

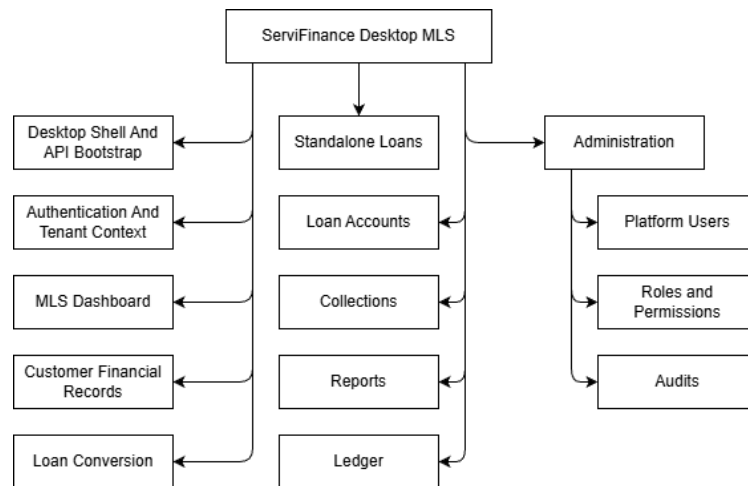
AuditEvents

Field	Data Type	Length / Precision	Description
Id	uniqueidentifier	36	Primary key of the audit event.
TenantId	uniqueidentifier	36	Tenant that owns the audit event.
Scope	nvarchar	50	Workspace scope such as Platform, SMS, or MLS.

Category	nvarchar	50	Audit category such as System or Security.
ActionType	nvarchar	100	Action name being audited.
Outcome	nvarchar	50	Result of the action.
ActorUserId	uniqueidentifier	36	Optional user who performed the action.
ActorName	nvarchar	200	Actor display name.
ActorEmail	nvarchar	50	Actor email address.
SubjectType	nvarchar	100	Type of record affected.
SubjectId	uniqueidentifier	36	Optional affected record ID.
SubjectLabel	nvarchar	300	Human-readable affected record label.
Detail	nvarchar	1000	Audit event detail.
IpAddress	nvarchar	80	Source IP address when available.
UserAgent	nvarchar	500	Client user agent when available.
OccurredAtUtc	datetime2	-	Date and time when the event occurred.

Module Diagram

Desktop MLS Module Diagram



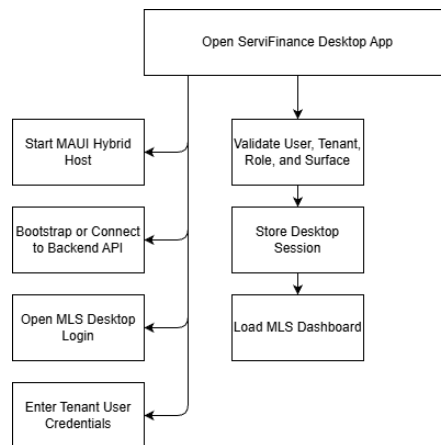
Module Responsibilities

Module	Main Responsibility
Desktop Shell And API Bootstrap	Starts the desktop experience and connects the MAUI host to the backend API.
Authentication And Tenant Context	Signs in tenant users and stores desktop session context.
MLS Dashboard	Shows finance overview, portfolio status, and important MLS metrics.
Customer Financial Records	Shows customer-level loans, balances, and ledger history.
Loan Conversion	Converts finance-ready service invoices into micro-loans.
Standalone Loans	Creates loans that do not originate from service invoices.
Loan Accounts	Reviews loan schedules, balances, payments, and reversible payment activity.
Collections	Shows due, overdue, and partially paid accounts for follow-up.

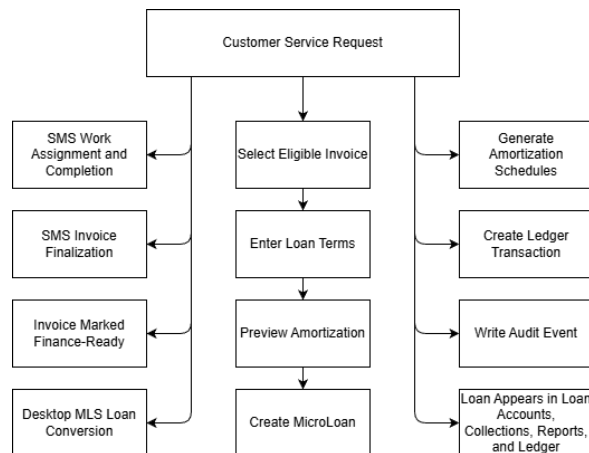
Reports	Shows finance and ledger reporting summaries.
Ledger	Shows transaction entries, balances, payment records, and finance trail.
Platform Users	Manages tenant users and assigns SMS/MLS platform roles.
Roles And Permissions	Manages available role behavior and permission matrix by platform scope.
Audits	Reviews MLS-related system and security events.

Process Flow Diagram

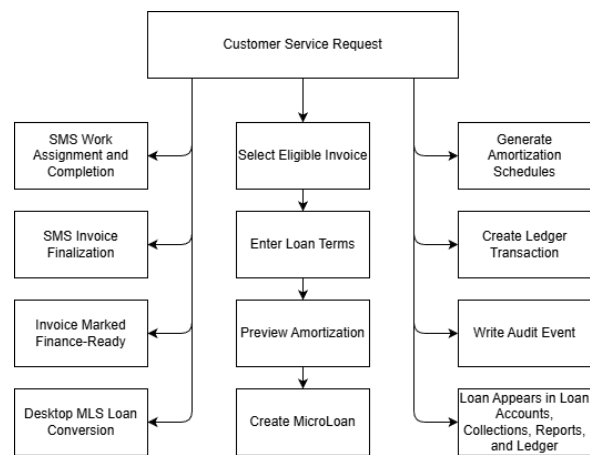
Desktop Access Flow



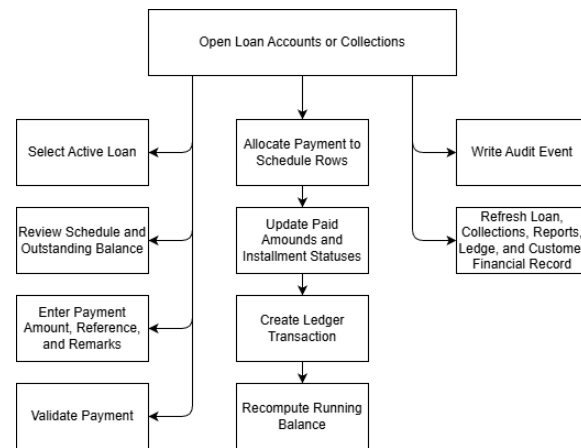
SMS Invoice to MLS Loan Flow



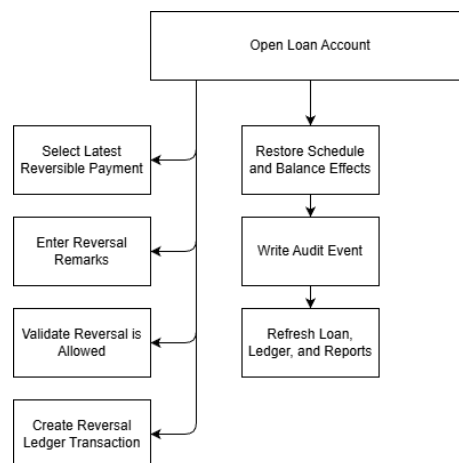
Standalone Loan Flow



Payment Posting Flow



Payment Reversal Flow



Reporting and Monitoring Flow

